Vision and Strategy: Regulating Fusion Machines Across the National Materials Program

March 11, 2025

ML25069A706

1. Vision

The U.S. fusion regulatory framework enables clear, efficient, independent, and reliable licensing and oversight through open processes.

The Principles of Good Regulation shape the approach to regulating the emerging fusion industry. Implementing this vision and strategy will design a focused approach suitable for commercial-scale fusion technologies.

- **Clear** – The NRC's decision on how to regulate near-term fusion technologies provides important clarity to the emerging fusion industry. Agreement States have had safely regulated fusion research and development systems under this framework for over 25 years, and the revised regulations and guidance will enable widespread implementation.
- **Efficient** – The byproduct materials framework enables a level of regulation that is commensurate with the hazards of a given application, while ensuring a high degree of technical competence to regulate fusion technologies. It is used to license technologies that range from portable gauges that measure the uniformity of highway pavement to major industrial facilities like irradiators for food and medical equipment. The National Materials Program (composed of the NRC and Agreement States) has demonstrated the flexibility to regulate emerging technologies such as cancer treatments and can do the same for fusion technologies.
- **Independent** – In making independent decisions on fusion—both the development of regulations and guidance and when reviewing specific applications—the U.S. regulatory community will openly seek all available facts and views. This includes coordination with other U.S. government agencies, Tribal governments, international regulators, academia, and fusion companies and non-government organizations and the public.
- **Reliable** – Fusion must be regulated in a way that maintains risks at an acceptably low level, considering uncertainties in technology development for components and a diversity of fusion designs. The regulatory framework will be designed in a way that lends stability to the planning process for commercialization of fusion and provides consistent decision-making on key safety and security questions. The National Materials Program (NMP) has extensive experience applying compatible requirements and guidance nationwide and will apply those principles to the regulation of fusion.
- **Open** – Extensive outreach is planned as the NRC and Agreement States develop the regulatory framework. This outreach will help other interested parties understand the approach, solicit feedback, and encourage consistent implementation across the National Materials Program.

2. Strategic Focus Areas

Three focus areas are needed to achieve the fusion vision: (1) **regulatory optimization**, (2) **technical readiness**, and (3) **partnership and coordination**.

2.1. Regulatory Optimization

Fusion regulation must optimize competing priorities of flexibility and consistency.

The byproduct materials framework is designed to provide a flexible and scalable approach—high-level performance-based regulations with tailored guidance to address specific technologies. The NRC's decision (see background section) to apply the byproduct materials framework was based on its understanding of the designs and hazards of the most likely near-term fusion applications. The regulations and guidance will be structured to account for the

known features of these designs. If designs introduce new hazards, materials, or other safety and security concerns, the regulator can require additional information or impose any necessary conditions to provide for adequate protection of public health and safety. Regulators will remain vigilant for hazards that go sufficiently beyond the near-term technologies and to recommend appropriate action.

The NRC's requirements in 10 CFR Part 30 and guidance in Volume 22 of NUREG-1556 will address scalability in situations when large quantities of tritium or activation products warrant additional actions. Some examples include:

- **Environmental reviews** – The environmental review process has the flexibility to determine the necessary level of review to support fusion licensing and will be dependent on the characteristics of the fusion application for the type of impacts across environmental resource areas. The NRC's environmental determination for several facility types, such as current research facilities using byproduct materials, is covered under a categorical exclusion under 10 CFR 51.22(c)(14).
- **Emergency preparedness** – Emergency plans and procedures are designed to be scaled to the potential hazards and accidents of each unique facility. Most applicants will not need an offsite emergency plan, while other applicants whose designs have significant inventories of tritium or activation products may need to implement offsite emergency plans.
- **Waste disposal** – Designs that only create radioactive waste that is well understood and classified in the existing NRC regulations will have a smooth disposal path; those that generate novel activation products will need to work with disposal facilities to ensure safe disposal, as well as evaluating security and other implications of these new materials.

With the exception of those compatibility areas where programs should be essentially identical, through the Commission's Agreement State Program Policy Statement and NRC's compatibility model, Agreement State radiation control programs have flexibility in program implementation and administration to accommodate individual State preferences, State legislative direction, and local needs and conditions. A State has the flexibility to design its own program, including incorporating more stringent, or similar, requirements provided that the requirements for adequate protection of public health and safety are met, and compatibility is maintained. The NRC will assess the compatibility of its fusion requirements and guidance as part of the rulemaking process.

Consistency is also important. In addition to Agreement State compatibility, current byproduct material approaches that provide consistency include safety evaluations that are shared across jurisdictions and nationwide registration of approved sealed sources and devices. The NRC will be exploring ways to provide consistent, predictable decision-making for fusion designs across the various National Materials Program jurisdictions to share design approvals of fusion machines designs for commercial distribution. This becomes especially important during the broader adoption as the NMP fusion regulatory program matures. This approach also supports jurisdictions whose regulators have not achieved their desired level of technical readiness.

2.2. Technical Readiness

Regulators need to strengthen fusion expertise as technology develops.

Currently, the wide variety of designs being considered in the marketplace and the lack of available design details at this early stage make it difficult for regulators to develop a deep understanding of all designs. This challenge will be alleviated to some extent as specific designs are proven and become more widely adopted. The NRC and Agreement States will hire and train staff to ensure they have the needed expertise to continue to license fusion technologies as they evolve. The NRC will partner with the Department of Energy (DOE), industry, academia, and international regulatory agencies to provide training to NRC and Agreement State staff on fusion confinement methods, fusion reactions, operation types, and design-specific safety considerations. Additional sessions by vendors will help provide a general baseline of understanding. The NRC anticipates taking the lead in offering licensing and inspection training, as is currently done for a variety of byproduct materials technologies, which will alleviate resource constraints on States and promote consistency. In addition, the NRC's Office of Nuclear Regulatory Research is evaluating the need for research on novel issues, especially those for materials effects, waste disposal and recycling, and computer codes for shielding calculations and offsite dose evaluations.

2.3. Partnership and Coordination

New designs cannot be regulated in a vacuum.

The NRC will continue close coordination with Agreement State partners to develop regulations, licensing guidance, inspection guidance, implementation strategies, and training. This includes State regulators' participation in working groups, discussion in government-to-government meetings, and formal comment on draft documents. In addition to providing training and regulatory infrastructure to enable States to review fusion applications, the NRC has established processes to directly support States in technical reviews through information sharing, user group, and technical assistance requests. The Integrated Materials Performance Evaluation Program enables the NRC and Agreement States to evaluate each other's technical readiness and implementation of fusion, as well as to offer best practices and insights for improvement.

The NRC will continue coordination with other Federal agencies with interests in fusion, including:

- **DOE** – DOE's Offices of Science and Fusion Energy Sciences supports significant research activities related to fusion, and the NRC can leverage these research activities to inform its regulatory approach. DOE, as it already does on advanced reactor issues, can partner with the NRC on outreach to interested parties, including environmental justice communities and Tribes, on the technology and regulatory approach.
- **National Nuclear Security Administration (NNSA)** – NNSA has interests in fusion associated with its nuclear nonproliferation mission. The NRC will continue to partner on appropriate regulatory requirements as well as design techniques and industry initiatives to address nonproliferation.
- **Department of Commerce** – The NRC is not proposing any changes to the current framework for export controls on tritium or other fusion associated isotopes. Under the current framework, the NRC coordinates with the Department of Commerce for export of materials under its jurisdiction.

The NRC is also committed to bilateral and multilateral cooperation with international regulatory counterparts. All benefit when regulators who are taking a similar byproduct materials approach share design and hazard insights, guidance documents, and operating experience. These

efforts are already underway with the United Kingdom and Canada and are likely to expand. The G7[1] has created a fusion energy working group to promote international collaborations to accelerate the demonstration of fusion potential, encourage private investment and public engagement, promote harmonization of fusion regulations, and share best practices. The International Atomic Energy Agency is also supporting the development of fusion technical reports and safety reports, as well as key elements for an overall fusion regulatory framework. The NRC is engaged in these workshops and conferences at both the staff and management levels.

The NRC is considering regulatory approaches to licensing standard fusion machine designs to facilitate commercial deployment. Regulators also coordinate with designers and applicants to understand designs, hazards, schedules, and other aspects critical to informing both strategy and actions. In addition, industry groups and standards organization are likely to provide valuable guidance to designers, as is already done in other byproduct materials areas (e.g., industry standards for irradiator design). Adherence to these third-party certifications may streamline the regulatory review process. As the NRC increases its resources applied to fusion regulation, it expects to increase this coordination.

3. Outcomes and Schedules

To achieve the vision, specific outcomes are needed related to each of the focus areas. These outcomes depend on the stage of maturity of the designs and associated regulatory needs. In each stage, outcomes are grouped into four objectives that drive NMP activities. These include:

- Objective 1: Establish regulations and guidance to ensure that fusion machines operate in a safe and secure manner;
- Objective 2: Develop a framework to license standard fusion machine designs for commercial deployment across the National Materials Program;
- Objective 3: Prepare NRC and Agreement State staff for efficient and effective fusion machine oversight and develop public trust in regulatory oversight of fusion machines; and
- Objective 4: Harmonize international efforts to embrace a risk-informed, scalable approach for fusion regulatory oversight.

The tables below show the schedule and sequence for outcomes in each stage, including overlaps driven by industry. For completed outcomes with publicly available documentation, they can be found at https://www.nrc.gov/materials/fusion-machine.html.

3.1. Stage 1: Regulatory Preparedness

Objective	Outcome	Schedule / Sequence	Lead Organization
1	Hold external engagement meetings on preliminary language / approach for rule and guidance	June 2023 – August 2024	NRC/NMSS
1	Standing Committee on Compatibility evaluate and assign compatibility designations on proposed rule and draft guidance	July 2024	NRC/NMSS Organization of Agreement States
1	Agreement State comment on proposed rule and draft guidance	August 2024	Organization of Agreement States
1	Deliver proposed rule and draft guidance to Commission	December 2024	NRC/NMSS

[1] The Group of Seven (G7) is an intergovernmental political and economic forum consisting of Canada, France, Germany, Italy, Japan, the United Kingdom and the United States.

Objective	Outcome	Schedule / Sequence	Lead Organization
1/3	Issue revision to proposed draft NUREG-1556, volume 22 licensing guidance	March 2025	NRC/NMSS
1/3	Publish proposed rule and draft guidance for public comment and host public meetings	2025	NRC/NMSS
1	Evaluate the appropriate environmental review process (i.e., development of an EIS, EA, or of a categorical exclusion category) for the expected range of fusion electrical generation stations based on expected footprint, need for transmission lines, etc.	2026	NRC/NMSS
1/3	Incorporate internal and public comments and issue final rule and licensing guidance	2027 (12/31/2027 legislative deadline – see background section)	NRC/NMSS
1/2	Participate in information-exchange meetings between RES and DOE, ARPA-E, EPRI, and standards development organizations	2023 - ongoing	NRC/RES NRC/NMSS
2	Provide any legislative proposals needed to facilitate licensing of mass-produced fusion machines	July 2024 (ADVANCE Act passage)	NRC/OGC
3	MIT-led fusion technology training seminar for NRC and Agreement State staff	November 2023	NRC/RES
3	Complete research white paper for longer-term fusion research and shorter-term technology familiarization	February 2024	NRC/RES
3	Complete skill gap assessment for licensing reviewer and inspector qualifications	2025	NRC/OCHCO
3	Hold fusion workshops for regulators	2025-2026	DOE Industry Academia
3	Hold knowledge-management sessions on specific fusion applications	2025-2027	Agreement State programs
4	Participate in IAEA working group on fusion safety and regulatory frameworks	May 2024	IAEA (NRC participation coordinated through OIP; DOE also participating)
4	Develop Fusion Key Elements and participate in IAEA World Fusion Energy Group	November 2024	IAEA (NRC participation coordinated through OIP; DOE also participating)

3.2. Stage 2: Prototypes, Demonstrations, and Other Single-Site Projects

Objective	Outcome	Schedule / Sequence	Lead Organization
1/3	Incorporate fusion program support in budget requests for NRC	2024 (for 2026 budget) - ongoing	NRC/NMSS
1	Determine level of inspection needed before licensing (beyond normal pre-licensing visits) or during construction of fusion facility (after licensing)	2025	NRC/NMSS
1	Review pre-licensing guidance to address any security concerns for fusion designs	2026 (needs more design information)	NRC/NMSS NRC/NSIR

Objective	Outcome	Schedule / Sequence	Lead Organization
1	Determine any licensing adjustments needed for complex fusion facilities and update guidance in NUREG-1556 if needed	Before ARC and Helion's first fusion power plant reviews complete	NRC/NMSS
1/3	Develop inspection procedure specific to fusion, adapting broad-scope and accelerator documents	2027 (operational stage)	NRC/NMSS
1	Determine if any additional guidance or reviews are needed to license offsite storage of tritium-containing fluids (e.g., for decay to helium-3)	2027	NRC/NMSS
3	Communicate process for NRC assistance with reviews of applications submitted to Agreement States	2024 - ongoing	NRC/NMSS
3	Complete white paper on guide for fusion licensing	2025	CRCPD
3	Conduct local outreach activities in areas where fusion designs will be built	2025 - ongoing	DOE NRC Industry
3	Review of additional commercial R&D/proof of concept facilities	2025 - ongoing	Agreement State programs
3	Establish fusion user group; create implementation toolkit and databank of safety evaluations for shared use on NRC website	2025	NRC/NMSS Agreement State programs
3	Hire appropriate level of staff to address fusion licensing support budget and conduct training as needed	2026-2027	NRC/NMSS NRC/Regions Agreement State programs
3	Complete review of Helion's first fusion power plant application	2026-2027	Washington Agreement State program
3	Complete review of Type One Energy's prototype facility application	2026-2027	Tennessee Department of Environment and Conservation
3	Complete review of Commonwealth Fusion Systems' ARC application	TBD	Virginia Agreement State program
3	Update accident codes as needed to enable confirmatory calculations; conduct other evaluations (engineering, risk) as needed	TBD (reference RES white paper)	NRC/RES
3	Conduct safety and environmental reviews for applications made to the NRC	TBD	NRC/NMSS
3	Complete review of Helion Polaris application	July 2024	Washington Agreement State program
3	Complete review of Commonwealth Fusion Systems SPARC application	October 2024	Massachusetts Agreement State program

3.3. Stage 3: Broader Adoption

Objective	Outcome	Schedule / Sequence	Lead Organization
1	Evaluate initial environmental reviews of Stage 2 applications and determine the appropriate environmental review	2027 - ongoing	NRC/NMSS

Objective	Outcome	Schedule / Sequence	Lead Organization
	process, building on other materials and advanced reactor reviews		
2	Develop industry readiness signposts and markers for implementing framework for commercial distribution of fusion machines	2025	NRC/NMSS NRC/Regions Agreement State programs
2	Issue ADVANCE Act Section 205 Report: Evaluate regulatory approaches that could enable efficient licensing of fusion machine designs for commercial distribution; obtain public comment on approaches as part of proposed rule	July 2025	NRC/NMSS NRC/Regions Agreement State programs
2	Issue industry standards, including ASME Division IV, on fusion machines	2027 - 2029	ASME
2	Evaluate the need to update guidance to endorse industry standards, as appropriate, and/or to credit third-party reviews of fusion designs, building on irradiator model	1 year after industry standards, if this approach emerges	NRC/NMSS
3	Complete RES “future focused” research projects relevant to fusion	TBD (reference RES white paper)	NRC/RES
3	Communicate to the Commission if any designs trigger thresholds in SRM direction; consider whether staff guidance or additional criteria are needed to facilitate this communication	As needed – not envisioned before the late 2030s based on industry discussions	NRC/NMSS
4	Issue international guidance on regulatory frameworks for fusion	2025-2026	IAEA

Appendix: Background

Fusion has been licensed at the research scale for decades, but updated tools and prepared regulators are needed to oversee this emerging industry effectively and efficiently.

Fusion systems harness the energy released in a controlled thermonuclear fusion reaction in which two nuclei combine to form a new nucleus. This process occurs in our Sun and other stars. Creating and sustaining the high temperature and pressure conditions for fusion on Earth has been a major technological challenge since the first controlled fusion in 1958. Significant progress has been made recently as technology advances and funding increases.

On January 3, 2018, the United States Congress passed the Nuclear Energy Innovation and Modernization Act (NEIMA) that requires the NRC to develop and implement the necessary regulatory frameworks for advance reactor designs by December 31, 2027. Fusion systems are included in the NEIMA's definition for advanced reactor. Therefore, the NRC is working to develop a clear and predictable regulatory framework for the near-term fusion system designs.

On January 3, 2023, the NRC staff submitted SECY-23-0001, "Options for Licensing and Regulating Fusion Energy Systems," summarizing fusion technologies, hazards, and regulatory approaches. The NRC staff provided three options for Commission consideration:

1. Categorization of fusion systems as utilization facilities with the NRC staff developing a new framework to address the associated specific hazards.
2. A byproduct material approach augmenting the framework in Title 10 of the *Code of Federal Regulations* (10 CFR), Part 30, “Rules of General Applicability to Domestic Licensing of Byproduct Material.”
3. A hybrid framework with decision criteria, based on the potential risks and hazards of a specific fusion system, to determine whether a byproduct material or a utilization facility approach is appropriate for that system.

On April 13, 2023, the Commission issued SRM-SECY-23-0001, "Staff Requirements – SECY-23-0001 – Options for Licensing and Regulating Fusion Energy Systems," approving the option for a limited-scope rulemaking to establish a regulatory framework for fusion systems that augments the NRC's byproduct material framework in 10 CFR Part 30. The Commission included additional direction for the staff:

- The staff should take into account the existence of fusion systems that already have been licensed and are being regulated by the Agreement States, as well as those that may be licensed prior to the completion of the rulemaking.
- The staff should develop a new volume of NUREG-1556, “Consolidated Guidance About Materials Licenses,” dedicated to fusion systems, so as to provide consistent guidance across the National Materials Program.
- The staff should evaluate whether controls-by-design approaches, export controls, or other controls are necessary for near-term fusion systems.
- If in the future, the staff, in consultation with the Agreement States, determines that an anticipated fusion design presents hazards sufficiently beyond those of near-term fusion technologies, the staff should notify the Commission and make recommendations for taking appropriate action as needed.

On March 7, 2024, NRC staff released the proposed draft version of NUREG-1556, Volume 22 “Consolidated Guidance About Materials Licenses Program-Specific Guidance About

Possession Licenses for Fusion Systems" (ML24067A227). This licensing guide would apply to fusion systems (now fusion machines) for research and development or commercial deployment. The guide was written to be technology neutral and could be used to license any of the fusion technologies be considered for commercial deployment. Volume 22 uses existing Part 30 requirements applicable to fusion machines in a manner like other Part 30 licensees as much as possible to integrate fusion machines into the existing byproduct material framework. The byproduct material frameworks used a scalable risk-informed approach that emphasizes containing, shielding, processing, or controlling radiation and radioactive materials that is amenable to the diversity of fusion machine technologies currently under development.

On July 9, 2024, the ADVANCE Act of 2024 was signed into law. Section 205 of the ADVANCE Act amended section 11 of the AEA to add the definition of "fusion machine" and amended the definition of "byproduct material" to include fusion machine generated radioactive material in section 11e.(3)(B). The ADVANCE Act also amended section 103 of NEIMA to delete "fusion reactor" and replace it with "fusion machine." Finally, subsection 205(c) of the ADVANCE Act requires the NRC to submit a report to Congress, by July 9, 2025, on design-specific licensing frameworks for "mass-manufactured fusion machines"; and provide an estimated timeframe for the NRC to issue regulations or guidance for licensing mass-manufactured fusion machines.

On December 11, 2024, the NRC staff submitted SECY-24-0085: "Proposed Rule: Regulatory Framework for Fusion Machines" (ML24019A060). This limited rulemaking includes requirements for content of application, environmental reviews, waste disposal, definitions, and other conforming changes to integrate fusion machines into the byproduct material framework. The proposed rule package also incorporated new and revised definitions from the ADVANCED Act. In addition to the proposed rule, a revised version of NUREG-1556, volume 22 is part of this rulemaking package.

The NRC continues to pursue this rulemaking in partnership with the Agreement States. In parallel, the NRC and Agreement States are developing the vision and strategy for regulating the full life cycle of fusion machines, informed both by materials licensing approaches and the NRC's vision and strategy for advanced reactors.

The NRC's Principles of Good Regulation—independence, openness, efficiency, clarity, and reliability—are embodied in this vision and strategy. While the NRC does not promote any technology, its responsibilities as a regulator include working effectively with all stakeholders, clearly communicating its requirements, and providing regulatory information and feedback in a timely manner. Above all, the NRC mission remains unchanged but the means to achieve its mission must be optimized. A National Materials Program that is strategically positioned to regulate fusion will provide necessary regulatory certainty to the fusion industry, potential applicants, and other stakeholders.